

EFFICIENT MITIGATION OF MALICIOUS INSIDER ATTACKS IN UNTRUSTED CLOUD SERVICES

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Abstract—Cloud computing is an evolving technology which has become an important portion of many organizations to store their data. It allows clients to outsource storage and computational resources on demand. Ensuring the privacy and security of the stored sensitive information is a major factor in the cloud computing environment. The existing cloud performs the sensitive data computation with the help of the Late Launch based technique called Flicker and Remote Attestation to verify the integrity of the cloud platform. Since the Flicker is a third party, it could not be completely trusted at all times. In this paper, we propose the Secure Virtual Machine (SVM) approach to ensure the confidentiality and integrity of client's sensitive data and computation on the cloud platform. The proposed work aims at providing a complete isolated and secured computation of security-sensitive data. Further, the Parallel Execution Technique helps in the simultaneous execution of both the security sensitive and normal computations thereby increasing the security and performance of the sensitive data computation efficiently.

Keywords—privacy, security, sensitive data, integrity, secured computation

I. INTRODUCTION

Cloud computing is a growing computing paradigm that offers many benefits to cloud users. This kind of computing paradigm enables convenient, on-demand network access to a shared pool of configurable computing resources, with the potential to reduce costs significantly through optimization and increasing operating and economics efficiencies [4]. Although cloud computing offers many benefits to cloud users, security remains a stumbling block in cloud adoption [5] [6] [7]. Insider threat represents one of the most significant concerns in the cloud computing environment [8]. This is owing to the impact it may have on the organization. An insider threat is regarded as any operation performed by legitimate users who can intentionally exceed the level of trust that is given to him/her by adversary affecting the confidentiality, integrity and availability of information stored in the cloud. Malicious insiders can betray the trust that is given to them by masquerading in the system and perform tasks that are not associated with their job roles. As a result, many businesses and individuals are not comfortable with the idea of moving their data and applications to systems they do not control. This necessitated the development of solutions that address the

problem of users masquerading in the system as normal users in order to carry out their malicious actions. One such solution is presented in this paper.

In current research work [22][23], on cloud platforms security has predominantly focused either on protecting these platforms from malicious cloud clients or on protecting cloud clients from each other's unwanted activities. The problem of protecting clients from the possible malicious acts of insiders such as cloud providers is not adequately addressed. There are organizations, for instance health-care and military, which are hesitant to move to cloud based services due to confidentiality concerns. Therefore practical solutions in this direction are required for wide adoption of cloud based services. In this paper, we propose an approach to ensure the confidentiality and integrity of client data and computation on the cloud platform. This is to ensure that private data is not exposed to internal parties such as the cloud administrator and other cloud clients.

The Secure Virtual Machine concept has been proposed. This architecture aims at fulfilling the integrity, confidentiality, security and isolation properties. The architecture provides a complete secure computation in cloud. It also includes various mechanisms and algorithms to provide integrity of data and maintains the security of the user's sensitive information. Our proposed work is to enhance the integrity of the cloud platform by securing the Virtual Machine (VM) launch operation and further preserving the confidentiality of client data by allowing the launched VM to perform operations on the sensitive data in full isolation.

II. RELATED WORK

Our work benefits from related work in trusted computing, trusted virtual machine monitors and specialized encryption based approaches. Several approaches in the first two categories are based on reliance on a trusted hypervisor also known as Trusted Virtual Machine Monitor (TVMM) [16][17]. The suggested TVMM model prevents platform administrators and other super users from examining or modifying client VMs. TVMM has the property of being able to defend its own integrity on the deployed platform. A typical VMM generally includes a hypervisor which in turn supports and executes multiple clients VMs. Hypervisors are

unfortunately complex software and as a result inflate the Trusted Computing Base (TCB). Due to hypervisors' complexities and large TCB there are many challenges that need to be addressed for the emergence of a practical TVMM.

Trusted computing base approaches

We further classify approaches that make use of a TCB into two generic categories, i.e. trusted virtual machine monitor and modified VMM.

Trusted virtual machine monitor based approaches

Terra [16] is one of the initial and influential research works that aims to provide secure closed box execution environment and an open general purpose system side by side on a single platform. Terra is based on a TVMM to make the function of a single node communicating in a distributed environment, transparent. This approach protects client computation integrity and confidentiality by providing a closed-box execution environment to a user VM from the platform administrator interference and inspection. The TVMM provides an interface to a management VM for allowing the allocation of memory, storage and other resources to the client VM. TVMM allows remote parties to trust the client VM closed-box execution environment by providing the attestation of the environment to the remote parties.

A technique for achieving client computational integrity and data confidentiality in an IaaS based cloud is presented in Khan et al. [18]. The technique is based on using remote attestation and a trusted virtual machine monitor (based on a TVMM). A trusted third party is used for the establishment of trust between cloud clients and cloud nodes. The cloud provider first attests and registers its nodes with a trusted third party which in turn verifies the cloud platform properties using remote attestation.

Modified VMM based approaches

CloudVisor [12], uses nested virtualization for preventing the administrative domain from encroaching upon the integrity and confidentiality of client VMs. CloudVisor divides the virtualization functionality into VM management and security protection. The management VM is responsible for management of client VMs while a tiny security monitor that runs beneath a commodity VMM such as Xen provides protection of client VMs integrity and confidentiality. Nested virtualization however imposes high overheads because privileged operations must be handled by both the bare-metal and nested hypervisor's, slowing down I/O intensive client applications.

Dewan et al. [19], present a technique to guard critical data of a client application running in a virtualized environment where a VM may include malware. The proposed approach uses a lightweight hypervisor for fine grained software-based run time memory protection. It places application critical data in protected memory regions and then registers it with the client application. Access to the protected memory region is controlled by the VMM on the basis of authenticity of an application.

Specialized encryption based approaches

Lei et al. [20], proposed homomorphic encryption techniques [21] for performing sensitive computation on encrypted data inside an untrusted cloud platform that provides input/output confidentiality and integrity. The authors have shown how common engineering and scientific computational tasks such as matrix inversion computation (MIC) can be securely outsourced to an untrusted cloud platform. The original matrix is encrypted before being outsourced to the cloud and then processed in encrypted form. Transformation is applied on the result received from the cloud to get accurate inversion of the original matrix. Homomorphic encryption, however limits the types of operations that can be performed. Moreover this approach overburdens the client and is not desirable for thin client environments, such as mobile devices, with limited resources.

A protocol for providing confidentiality and controlling access to the data in the cloud is proposed by Tysowski et al. [10]. The given approach recommends modifications to attribute-based encryption to control access to data based on possession of certain attributes by authorized users. The data owner determines attributes required for data access to protect data confidentiality from unauthorized users. The data owner then generates, encrypts and then uploads data to the cloud to be accessed only by authorized users.

The goal is to show that how the malicious acts of insiders that masquerade in the system can be reduced using our approach. As an effort to contribute a solution to a problem which is causing businesses and individuals to be reluctant about adopting the cloud, we propose a model that reduces attacks that originate from the legitimate users of the cloud such as cloud clients, cloud service providers and cloud system admin.

III. OUR APPROACH

A. Architecture

Fig. 1 shows our proposed architecture of the cloud environment to check the confidentiality and integrity of client data. There are four main actors – User, Cloud Service Provider, Node Controller and Secure Virtual Machine.

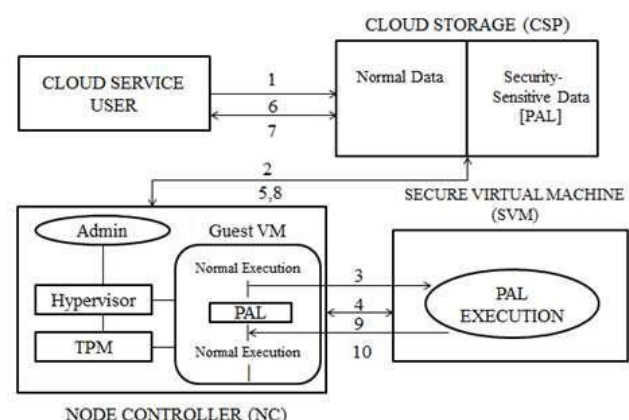


Fig. 1. System architecture

The working of the proposed architecture is as follows:

Step 1: Users register with their details. The normal and sensitive data are stored in Cloud Storage (CS) on the cloud while the sensitive data - Piece of Application Logic (PAL) need to be stored in an encrypted form.

Step 2: Both the types of data are sent to Node Controller (NC). The Node Controller executes only the normal data.

Step 3: The sensitive data is simultaneously executed in the Secure Virtual Machine (SVM).

Step 4: SVM then asks for the Decryption Key (Dk) of the client to decrypt the sensitive data.

Step 5: The NC sends the key request to Cloud Storage of Cloud Service Provider (CSP).

Step 6: The CSP then makes the request for the Decryption Key to the client.

Step 7: The client shares the Decryption Key to the CSP.

Step 8: And the CSP sends the key back to the Node Controller.

Step 9: Node Controller directs the Decryption Key (Dk) to the Secure Virtual Machine and then the sensitive data is decrypted using the key and executed in isolation.

Step 10: Finally, the security sensitive data computation result is then sent to Node Controller and used for computation of the normal data.

B. Secure VM Launch Protocol

The client VM is stored in an encrypted form on the Cloud Storage, so that it can only be launched on trusted NCs. The goal of this protocol is to retrieve the VM decryption key Dk_{VM} securely from the client, decrypt the SVM and then launch it on a trusted cloud node.

C. Confidentiality Sensitive Computation Protocol

The computation inside the security sensitive client VM is divided into two categories, normal computation and security sensitive computation. The normal computation takes place as usual on the virtualized platform with the system administrator running in a special administrative VM, which runs and controls client guest VMs. This special admin VM runs under the control of a platform owner. The same procedure however, cannot be followed for security sensitive computation as an insider such as the system administrator can run arbitrary processes in special admin VM and can compromise the confidentiality of client data and computation. The integrity information exchanged during VM launch cannot guarantee this protection.

In our proposed protocol the computation on sensitive data is organized as PAL, which executes inside SVM protection. The sensitive data is only visible to the client PAL inside the SVM session and is processed there in full isolation from the rest of the system. In this way, confidentiality of client data is enforced. The client can perform sensitive computation on the

cloud node in full isolation from the underlying NC software stack. Thus, client confidentiality is ensured, protecting the data from other guest VMs and the NC system administrator.

D. Remote Attestation

In remote attestation, the platform (firmware and software) configuration is captured and stored in a tamper resistant and cost effective chip called a Trusted Platform Module (TPM). A TPM is a secure storage area where cryptographic keys and other secure data can be stored. The key and data stored inside the TPM is protected from malicious alteration. The data stored inside the TPM normally includes platform configuration status. The platform status stored inside TPM can then be provided to external entities to convey the platform trustworthiness. Confidential information is held inside the TPM and is then signed and reported to a remote entity for verification and attestation purposes. This entire process is termed as remote attestation by the Trusted Computing Group (TCG) [24].

In remote attestation, to the TPM chip some form of integrity measurement system such as Linux Integrity Measurement Architecture (IMA) [25], is needed to generate and report the attestation of the system to the remote entity. TPMs [24], store platform integrity in the form of hashes of loaded software in data registers called Platform Configuration Registers (PCRs). Quote operation is used to attest the values of TPM PCRs. TPM Quote comprises of a subset of PCRs values together with a nonce all signed by a TPM Endorsement Key (EK). The private part of the EK is used for signing purposes during Quote generation and is used to convince remote verifiers that assertions in the TPM Quote have been signed by a trusted TPM.

E. Secure Virtual Machine

In existing system, Flicker is used to check the integrity and secure execution of sensitive data. But this Flicker is a semi-trusted infrastructure model since it may get compromised with a hacker and also has cost and performance issues. So in our proposed system, this semi-trusted Flicker is ignored. Thus, Secure Virtual Machine (SVM) has been proposed to execute the sensitive files in full isolation and to provide integrity of data while maintaining the privacy of the user. The architecture also aims to achieve the trust preserving VM launch operation.

In the proposed system, a user wants to share the data on the cloud. The users have to register themselves with their information in the cloud. After receiving it, the confidentiality sensitive information i.e. PAL is stored in a trusted NC on cloud by applying secure VM launch protocol. The NC passes the sensitive data to the SVM. The SVM then requests the client which provides the decryption key. The file gets decrypted and it is executed in the SVM in isolation. Then the computation result of the sensitive file is uploaded in the cloud. So again if the user needs to perform any modifications on the data stored in the cloud, the computation will be carried out in SVM securely and in isolation again. The dynamic operations

such as add, delete, modify can be performed on the data stored in the NC on cloud. The reverse process takes place for retrieving and modifying the file. Security sensitive computation is done by the proposed system by preserving the integrity and confidentiality of the user's data and maintaining the secrecy of the sensitive information. The Secure Virtual Machine utilizes the information stored in NC to perform computation operations. This reduces the computation cost. Virtual Machines are used to store and execute the user files. The information stored here cannot be hacked, leaked or attacked by unauthorized users and malicious insiders.

F. Parallel Execution Model

The Flicker session suspends the normal execution of a cloud node and runs in full isolation. This can be considered as a type of financial loss to the platform owner because the platform resources are not fully and efficiently utilized. Separate execution of security sensitive and normal computations on different cloud nodes decreases the performance.

Fig. 2 shows the Parallel Execution Model. In order to avoid the suspension of normal execution on a node, a Parallel Execution Model has been proposed to have only a subset of CPU cores assigned to a SVM session while normal execution continues on other cores. This way, execution of security sensitive and normal computations will take place at the same time side by side on the same cloud node. The proposed work increases the performance efficiently as well as maintains the security of the sensitive data computation and also incurs optimal cost. Hence, it ensures that the cloud platform owner or the cloud administrator is not compromising the integrity and confidentiality of client's security sensitive data and computation.

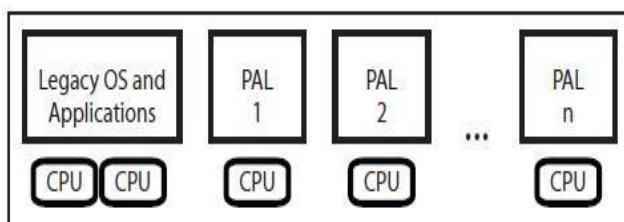


Fig. 2. Parallel execution model

In proposed system, Advanced Encryption Standard (AES) with a key size of 128-bits is used for encryption and decryption of the sensitive file. The encrypted file outsourced by the user is stored in the Node Controller on the cloud and decrypted by the user when the file is executed in the Secure Virtual Machine.

IV. EXPERIMENTAL RESULTS

A. Performance Evaluation and Comparison of Existing and Proposed Algorithms (RSA and AES)

1) Experimental Design

Four text files of different sizes are used to conduct four experiments, by which a comparison of the two algorithms

AES and RSA is performed. Performances of both the algorithms are evaluated considering the following four parameters:

- Encryption Time
- Decryption Time
- Key Generation Time
- Execution Time

2) Experimental Results and Analysis

a) Encryption Time

Fig. 3 shows the encryption time of AES and RSA on different file sizes. It is clear from the figure that encryption time of AES algorithm is less than RSA. Encryption algorithm plays a very important role in communication security. The encryption time is considered the time that an encryption algorithm takes to produce a cipher text from a plain text. Encryption time is used to determine the throughput of an encryption scheme, which is calculated as the total plaintext in bytes encrypted divided by the encryption time. Comparison analyses of the results of the selected encryption schemes are performed. Based on the text files used and the experimental result, it was concluded that AES algorithm consumes lesser encryption time than RSA.

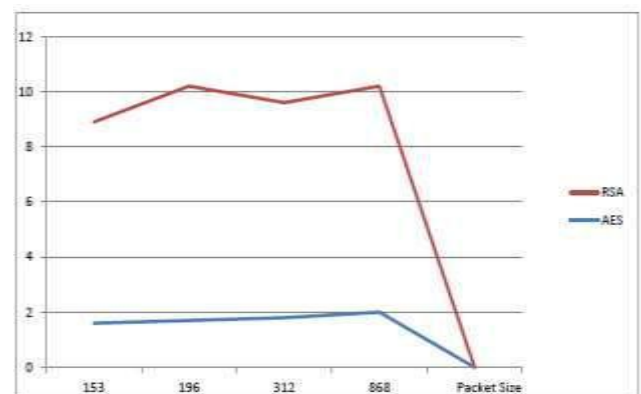


Fig. 3. Experimental result of RSA and AES encryption time

b) Decryption Time

Fig. 4 shows the decryption time of AES and RSA on different file sizes. The performance results conclude that the decryption time of RSA algorithm is more than AES algorithm.

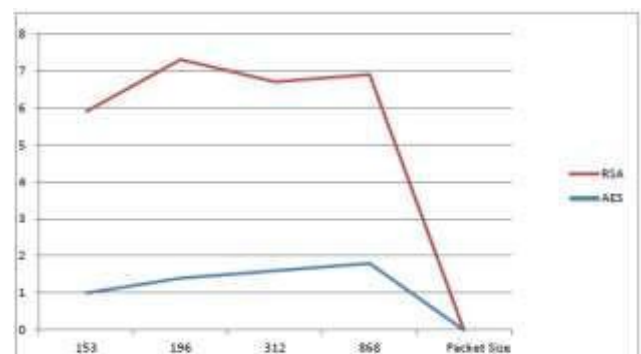


Fig. 4. Experimental result of RSA and AES decryption time

c) Key Generation Time

Fig. 5 shows the key generation time of symmetric AES and asymmetric RSA algorithms. Key generation time depends on the bit length of a key. Greater in length, more is the time. The RSA algorithm takes more time to generate the key because of key length of 1024 bits while AES algorithm takes less time because of key length of 128 bits.

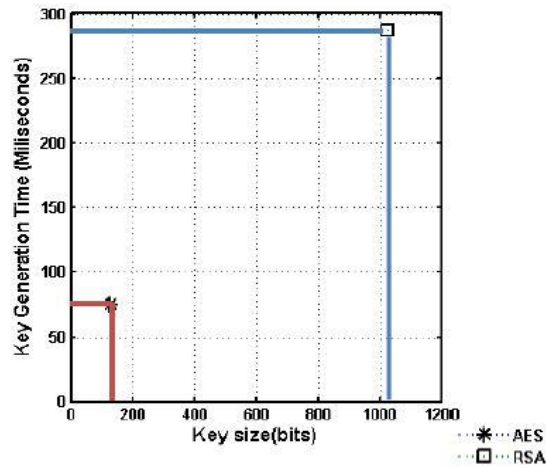


Fig. 5. Experimental result of RSA and AES key generation time

d) Execution Time

Fig. 6 shows the execution time of AES and RSA algorithms. It shows that the execution time calculated for the proposed AES algorithm is lesser, much faster and efficient than the execution time of the existing RSA algorithm.

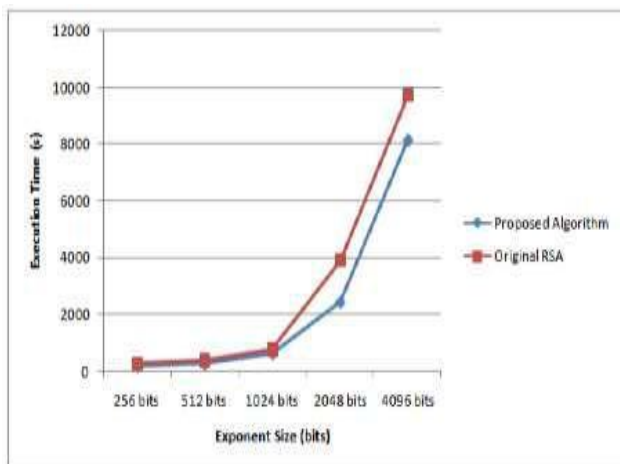


Fig. 6. Experimental result of RSA and AES execution time

Experimental results in Table I shows the Encryption and Decryption Times of RSA and AES algorithms with their different file sizes. Performance results show that when we increase the size of text files, the encryption and decryption time is also increased. By analyzing Table I, Time taken by AES algorithm for both encryption and decryption process is smaller compared to the time taken by RSA algorithm.

Experimental results in Table II shows the Key Generation Times of both RSA and AES algorithms with their key sizes. Comparison results show that the AES algorithm takes less time to generate the key while RSA algorithm takes more time. From these experimental results, it has been observed that the proposed AES algorithm is much better and efficient than the existing RSA algorithm.

TABLE I. RSA AND AES ENCRYPTION AND DECRYPTION TIMES ALONG WITH FILE SIZES

S.NO	Algorithm	Packet/File Size (KB)	Encryption Time (Sec)	Decryption Time (Sec)
1	RSA	153	7.3	4.9
	AES		1.6	1
2	RSA	196	8.5	5.9
	AES		1.7	1.4
3	RSA	312	7.8	5.1
	AES		1.8	1.6
4	RSA	868	8.2	5.1
	AES		2.0	1.8

TABLE II. RSA AND AES KEY SIZES WITH THEIR KEY GENERATION TIMES

Cryptographic Algorithms		Key Size (bits)	Key Generation Time (millisecond)
Asymmetric	RSA	1024	287
Symmetric	AES	128	75

V. CONCLUSION

The proposed architecture overcomes the drawbacks of the existing system and also helps to maintain the efficiency of the system while achieving the objectives. The architecture provides complete prevention, reduction and isolation from malicious insider attacks by preserving user's privacy and maintaining the confidentiality and integrity of the data. It also provides a high level of security and performance.

The Secure Virtual Machine executes the security-sensitive data and code in full-isolation. Since the virtual machines are kept separately, there is no means for anybody to access it. Also the users do not have any knowledge of the data executed on them. Since sensitive data are kept very confidential, the privacy of the user is preserved. AES encryption/decryption

encrypts the data before storing in the cloud and decrypts them before it is executed on the virtual machine. The architecture utilizes information stored in the cloud and provides complete isolated execution of the user's data upon request. There is no chance for any malicious intrusion or malpractice because this Secure Virtual Machine runs separately and all security-sensitive data must go through this architecture which provides security during access and computation.

Further, to facilitate parallel execution of all types of data, a Parallel Execution Technique has been proposed that provides simultaneous execution of security sensitive and normal computations at the same time side by side on the same cloud node thereby increasing the security and performance of the sensitive data computation efficiently. It has been concluded that the data integrity, confidentiality and performance have been obtained with a high level of security and optimal cost thereby, also executing the security-sensitive codes with strong and complete isolation. Thus, the proposed architecture fulfills all the objectives designed for the proposed system and also overcomes the drawbacks of existing techniques. There are also certain limitations of the proposed architecture which will be considered as a future enhancement of the project such as extending our system by using multiple CSPs for data storage purpose.

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